

2D Maximization solved w/ Graphical

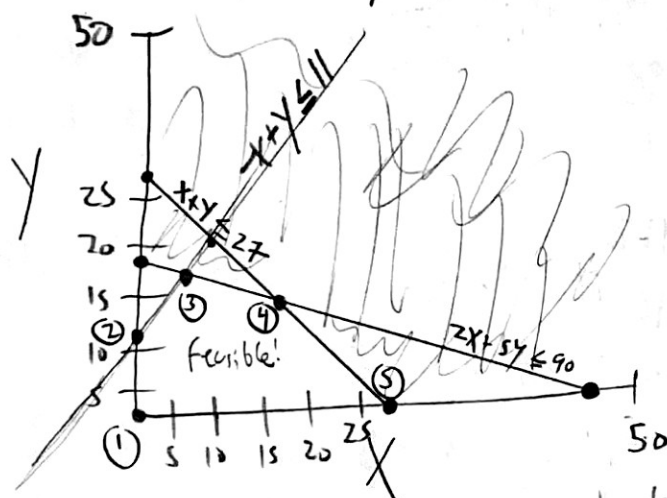
(1)

$$\text{Max}(z) = 4X + 6Y$$

$$1) -X + Y \leq 11$$

$$2) X + Y \leq 27 \quad X, Y \geq 0$$

$$3) 2X + 5Y \leq 90$$



$$\text{Eqn 1: } -X + Y \leq 11$$

$$\text{when } X = 0, Y = 11$$

$$\text{when } Y = 0, X = -11$$

* need to extend! try a value on X...

$$(0, 11) \text{ and } (-11, 0) \text{ and } (\underline{10}, 21)$$

$$\text{when } \underline{X=10}, Y=21$$

$$\text{Eqn 2: } X + Y \leq 27$$

$$\text{when } X = 0, Y = 27$$

$$\text{when } Y = 0, X = 27$$

$$(0, 27) \text{ and } (27, 0)$$

$$\text{Eqn 3: } 2X + 5Y \leq 90$$

$$(0, 18) \text{ and } (45, 0)$$

$$\text{when } X = 0, Y = \frac{90}{5} = 18$$

$$\text{when } Y = 0, X = \frac{90}{2} = 45$$

these 3 constraints make a feasible region w/ five corner points.

Some CPF points are obvious, others are tough to read.

(2)

• CP 1 = (0, 0) • CP 3 = ? • CP 5 = (0, 27)

• CP 2 = (0, 11) • CP 4 = ?

→ To solve for CP 3, identify which constraints are intersecting.

$$\begin{array}{rcl} 2X + 5Y & = & 90 \\ 2(-X + Y) & = & 11 \end{array}$$

• cancel out one variable, solve, then plug and chug for other

$$7Y = 112$$

$$Y = 16$$

$$\boxed{CP\ 3 = (5, 16)}$$

now plug and chug → $-X + 16 = 11$
 $X = 5$

→ To solve for CP 4, identify which constraints are intersecting

$$\begin{array}{rcl} 2X + 5Y & = & 90 \\ X + Y & = & 27 \end{array} \quad \leftarrow \text{multiply } X2 \text{ and subtract}$$

$$2X + 5Y - 2X - 2Y = 90 - 54$$

$$3Y = 36$$

$$Y = 12 \quad \text{so } X = 15$$

$$\boxed{CP\ 4 = (15, 12)}$$

→ now that CPFs are identified, evaluate objective function!

- CP 1 = (0, 0) = $4(0) + 6(0) = 0$
- CP 2 = (0, 11) = $4(0) + 6(11) = 66$
- CP 3 = (5, 16) = $4(5) + 6(16) = 116$
- CP 4 = (15, 12) = $4(15) + 6(12) = 132$
- CP 5 = (0, 27) = $4(0) + 6(27) = 162$

2D Simplex Maximization Problem

①

$$\max (z) = 4x + 6y$$

$$\text{subject to: } -x + y \leq 11$$

$$x + y \leq 27$$

$$x, y \geq 0$$

$$2x + 5y \leq 90$$

→ now let's write the augmented form of the model

$$\max (z) = 4x + 6y$$

$$-x + y + s_1 = 11$$

$$x + y + s_2 = 27$$

$$2x + 5y + s_3 = 90$$

$$x, y, s_1, s_2, s_3 \geq 0 \text{ (non-negativity)}$$

Iteration #1

→ write the initial simplex tableau

	X	Y	s_1	s_2	s_3	RHS
R_0	-1	1	1	0	0	11
R_1	1	1	0	1	0	27
R_2	2	5	0	0	1	90
R_3	-4	-6	0	0	0	0

why -4 and -6? $z = 4x + 6y$ so $-4x - 6y = z$
and we know simplex chooses origin as initial soln $(0,0)$

→ what's the most negative number? -6
So, Y is the entering basic variable

→ use minimum ratio test to determine which is leaving basic variable.

$$R_0 = 11/1 = 11 \quad R_1 = 27/1 = 27 \quad R_2 = 90/5 = 18$$

So select R_0 which corresponds to s_1 .

we already have a "1" in the pivot, so now we do row operations (GJ elimination - for readability)

(2)

→ multiply (divide by a constant)

→ add (subtract) multiples of a row to another

• $R_0^* = R_0$ which gives $(-1, 1, 1, 0, 0, 11)$

• $R_1^* = R_1 - R_0^*$

R_1	1	1	0	1	0	27
$- R_0^*$	-1	1	1	0	0	11
R_1^*	2	0	-1	1	0	16

• $R_2^* = R_2 - 5R_0^*$

R_2	2	5	0	0	1	90
$- 5R_0^*$	-5	5	5	0	0	55
$- R_2^*$	7	0	-5	0	1	35

• $R_3^* = R_3 + 6R_0^*$

R_3	-4	-6	0	0	0	0
$+ 6R_0^*$	-6	6	6	0	0	66
R_3^*	-10	0	6	0	0	66

Iteration #2

→ now we can rewrite the tableau

	X	Y	S_1	S_2	S_3	RHS
R_0	-1	1	1	0	0	11
R_1	2	0	-1	1	0	16
R_2	7	0	-5	0	1	35
R_3	-10	0	6	0	0	66

optimality test: since there is a negative value in bottom row, we keep pivoting.

X is entering basic variable.

• R_0 is infeasible (neg ratio)

• $R_1 = \frac{16}{2} = 8$

• $R_2 = 35/7 = 5 \leftarrow \text{minimum!}$

• R_3 is infeasible (neg ratio)

use row operations...

• $R_2^* = \frac{1}{7} R_2$ (gives "1" in the pivot element)

• $R_0^* = R_0 + R_2^*$

• $R_3^* = R_3 + 10R_2^*$

• $R_1^* = R_1 - 2R_2^*$

③

⇒ You can do mental math now that you have demonstrated competence...

*

	X	Y	S_1	S_2	S_3	RHS
R_0	0	1	$\frac{2}{7}$	0	$\frac{1}{7}$	16
R_1	0	0	$\frac{3}{7}$	1	$-\frac{2}{7}$	6
R_2^*	1	0	$-\frac{5}{7}$	0	$\frac{1}{7}$	5
R_3	0	0	$-\frac{8}{7}$	0	$\frac{10}{7}$	116

Iteration #3 $6 + \left(-\frac{50}{7}\right) = \frac{42}{7} - \frac{50}{7} = -\frac{8}{7}$

There is STILL a negative number in the bottom row...
PIVOT AGAIN! Before we do...

We can show the row operations for completeness...

$R_0^* = R_0 + R_2^*$

R_0	-1	1	1	0	0	11
R_2^*	1	0	$-\frac{5}{7}$	0	$\frac{1}{7}$	5
R_0^*	0	1	$\frac{2}{7}$	0	$\frac{1}{7}$	16

$R_1^* = R_1 - 2R_2^*$

R_1	2	0	1	1	0	16
$2R_2^*$	2	0	$-\frac{10}{7}$	0	$\frac{2}{7}$	10
R_1^*	0	0	$\frac{3}{7}$	1	$-\frac{2}{7}$	6

$R_3^* = R_3 + 10R_2^*$

R_3	-10	0	6	0	0	66
$10R_2^*$	10	0	$-\frac{50}{7}$	0	$\frac{10}{7}$	50
	0	0	$-\frac{8}{7}$	0	$\frac{10}{7}$	116

$\frac{42}{7} - \frac{50}{7} = -\frac{8}{7}$

* (a little cleaner to use a table!)

error in book!
this $-\frac{2}{7}$ is correct!

Pivot 3 there's a $-\frac{8}{7}$ in the bottom row,
so we keep going.

(4)

The minimum ratio test is MAXN.

$$R_0 = 16 \times \frac{7}{2} = 56$$

$$R_1 = 6 \times \frac{7}{3} = 14 \rightarrow \text{winner! minimum}$$

R_2 and R_3 have negative numbers so they aren't considered for minimum ratio test.

$\rightarrow$ the row operations are:

$$R_1^* = \frac{7}{3} R_1 \text{ which gives } (0, 0, 1, \frac{7}{3}, -\frac{2}{3}, 14)$$

$$R_0^* = R_0 - \frac{2}{7} R_1^*$$

$$R_2^* = R_2 + \frac{5}{7} R_1^*$$

$$R_3^* = R_3 + \frac{8}{7} R_1^*$$

R_0	0	1	$\frac{2}{7}$	0	$\frac{1}{7}$	16
$-\frac{2}{7} R_1^*$	0	0	$\frac{2}{7}$	$\frac{2}{3}$	$-\frac{4}{21}$	4
R_0^*	0	1	0	$-\frac{2}{3}$	$\frac{1}{3}$	12

$$\frac{3}{21} + \frac{4}{21} = \frac{7}{21} = \frac{1}{3}$$

R_2	1	0	$-\frac{5}{7}$	0	$\frac{1}{7}$	5
$+\frac{5}{7} R_1$	0	0	$\frac{5}{7}$	$\frac{5}{3}$	$-\frac{10}{21}$	10
R_2^*	1	0	0	$\frac{5}{3}$	$-\frac{1}{3}$	15

$$\frac{5}{7} \times \frac{7}{3} = \frac{35}{21} = \frac{5}{3} \quad \frac{5}{7} \times 14 = 10$$

$$\frac{5}{7} \times -\frac{2}{3} = -\frac{10}{21}$$

$$\frac{3}{21} - \frac{10}{21} = -\frac{7}{21} = -\frac{1}{3}$$

R_3	0	0	$-\frac{8}{7}$	0	$\frac{10}{7}$	116
$+\frac{8}{7} R_1^*$	0	0	$\frac{8}{7}$	$\frac{8}{3}$	$-\frac{16}{21}$	16
R_3^*	0	0	0	$\frac{8}{3}$	$\frac{2}{3}$	132

$$\frac{8}{7} \times \frac{7}{3} = \frac{8}{3}$$

$$\frac{8}{7} \times 14 = 16$$

$$\frac{8}{7} \times -\frac{2}{3} = -\frac{16}{21}$$

$$\frac{30}{21} - \frac{16}{21} = \frac{14}{21} = \frac{2}{3}$$

→ let's write the final tableau and see if we're done (5)

	X	Y	S_1	S_2	S_3	RHS
R_0	0	1	0	$-2/3$	$1/3$	12
R_1	0	0	1	$7/3$	$-2/3$	14
R_2	1	0	0	$5/3$	$-1/3$	15
R_3	0	0	0	$8/3$	$2/3$	132

no negative numbers in bottom row means we are done!

$$(X, Y, S_1, S_2, S_3) = (15, 12, 14, 0, 0)$$

$$\text{and } Z = 132$$

this basic feasible solution is optimal.